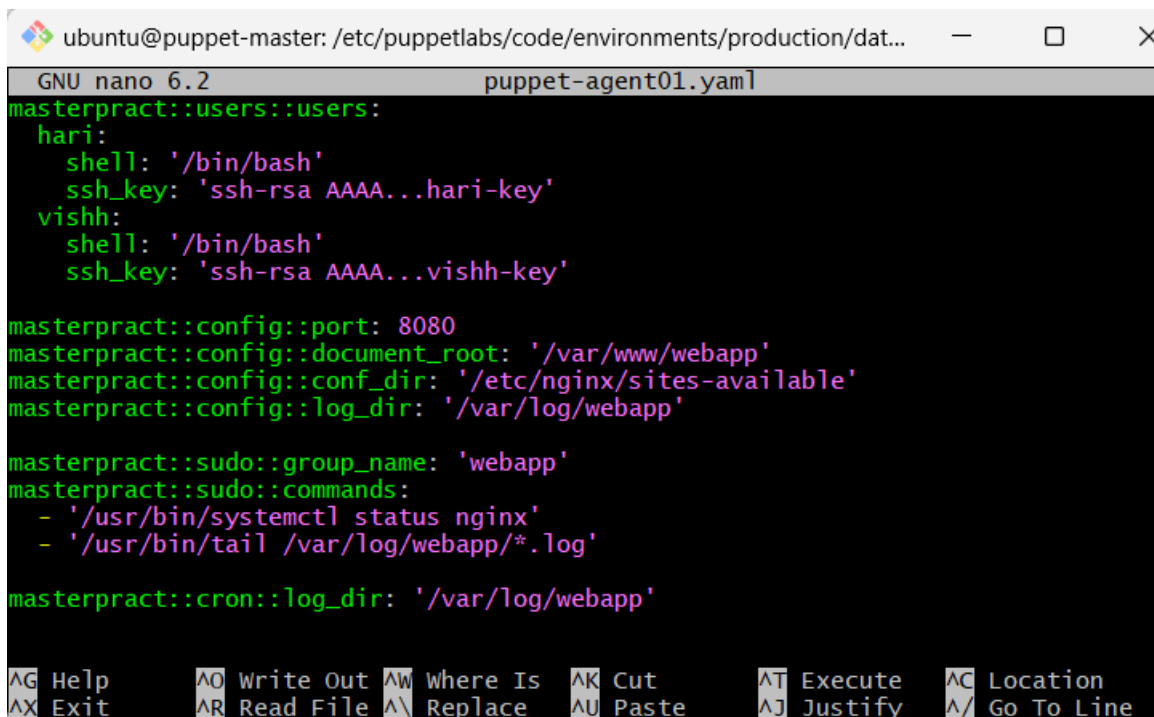


Puppet-Mastery Part-12(Special Armageddon Master Practical-2) Configuration-Driven Infrastructure (user creation, Cron, Sudo configs) and Web Application Deployment with Puppet, Hiera, Facts, EPP Templates, Custom Functions and Nginx

Master:



```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
GNU nano 6.2 puppet-agent01.yaml
masterpract::users::users:
  hari:
    shell: '/bin/bash'
    ssh_key: 'ssh-rsa AAAA...hari-key'
  vishh:
    shell: '/bin/bash'
    ssh_key: 'ssh-rsa AAAA...vishh-key'

masterpract::config::port: 8080
masterpract::config::document_root: '/var/www/webapp'
masterpract::config::conf_dir: '/etc/nginx/sites-available'
masterpract::config::log_dir: '/var/log/webapp'

masterpract::sudo::group_name: 'webapp'
masterpract::sudo::commands:
  - '/usr/bin/systemctl status nginx'
  - '/usr/bin/tail /var/log/webapp/*.log'

masterpract::cron::log_dir: '/var/log/webapp'

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Hiera Configuration used for this practical

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/manifests$ sudo cat init.pp
class masterpract{

    contain masterpract::install
    contain masterpract::config
    contain masterpract::users
    contain masterpract::service
    contain masterpract::sudo
    contain masterpract::cron

    Class['masterpract::install']->Class['masterpract::config']->
    Class['masterpract::service']

    Class['masterpract::users']->Class['masterpract::config']

    Class['masterpract::users']->Class['masterpract::sudo']
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/manifests$ sudo cat config.pp
class masterpract::config(
    Integer $port,
    String $document_root,
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/manifests$ sudo cat config.pp
class masterpract::config(
    Integer $port,
    String $document_root,
    String $conf_dir,
    String $log_dir
){

    file{$log_dir:
        ensure=>directory,
        owner=>'www-data',
        group=>'www-data',
        mode=>'0750',
    }

    file{'/etc/nginx/sites-enabled/default-000.conf':
        ensure=>absent
    }

    file{"${conf_dir}/nginx.conf":
        ensure=>file,
        owner=>'webapp',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
file{"${conf_dir}/nginx.conf":
  ensure=>file,
  owner=>'webapp',
  group=>'webapp',
  mode=>'0440',
  content=>epp('masterpract/nginx.conf.epp',
    {
      'document_root'=>$document_root,
      'log_dir'=>$log_dir,
      'port'=>$port
    }
  ),
  notify=>Service['nginx']
}

file['/etc/nginx/sites-enabled/vishsite':
  ensure=>link,
  target=>"${conf_dir}/nginx.conf",
  notify=>Service['nginx'],
  require=>File["${conf_dir}/nginx.conf"]
}

file{$document_root:
  ensure=>directory,
  owner=>'webapp',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
file{$document_root:
  ensure=>directory,
  owner=>'webapp',
  group=>'webapp',
  mode=>'0755'
}

file{"${document_root}/index.html":
  ensure=>file,
  owner=>'webapp',
  group=>'webapp',
  mode=>'0644',
  content=>epp('masterpract/index.html.epp',
    {
      'document_root'=>$document_root
    }
  ),
  require=>File[$document_root]
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master
pract/manifests$ sudo cat service.pp
class masterpract::service{

  service{'nginx':
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
pract/manifests$ sudo cat service.pp  
class masterpract::service{  
  
  service{'nginx':  
    ensure=>running,  
    enable=>true  
  }  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/manifests$ sudo cat users.pp  
class masterpract::users(  
  String $group_name='webapp',  
  Hash $users  
) {  
  
  group{$group_name:  
    ensure=>present  
  }  
  
  $users.each |String $username,Hash $user_data| {  
  
    $home = masterpract::user_home($username)  
  
    user{$username:  

```

```
$home = masterpract::user_home($username)  
  
    user{$username:  
      ensure=>present,  
      gid=>$group_name,  
      home=>$home,  
      shell=>$user_data['shell'],  
      managehome=>true,  
      require=>Group[$group_name],  
    }  
  
    file{"${home}/.ssh":  
      ensure=>directory,  
      owner=>$username,  
      group=>$group_name,  
      mode=>'0700',  
      require=>User[$username]  
    }  
  
    file{"${home}/.ssh/authorized_keys":  
      ensure=>file,  
      owner=>$username,  
      group=>$group_name,  

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
file{"${home}/.ssh/authorized_keys":  
  ensure=>file,  
  owner=>$username,  
  group=>$group_name,  
  mode=>'0600',  
  content=>$user_data['ssh_key'],  
  require=>File["${home}/.ssh"]  
}  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/manifests$ sudo cat cron.pp  
class masterpract::cron(  
  String $log_dir  
)  
{  
  
  cron{'log-cleanup':  
    ensure=>present,  
    user=>'www-data',  
    minute=>'0',  
    hour=>'2',  
    month=>'*',  
    monthday=>'*',  

```

```
monthday=>'*',  
weekday=>'*',  
command=>"/usr/bin/find ${log_dir} -type f -mtime +7 -delete",  
require=>File[$log_dir]  
}  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/manifests$ sudo cat sudo.pp  
class masterpract::sudo(  
  String $group_name='webapp',  
  Array[String] $commands  
)  
{  
  
  file{"/etc/sudoers.d/${group_name}":  
    ensure=>file,  
    owner=>'root',  
    group=>'root',  
    mode=>'0440',  
    content=>epp('masterpract/sudoers.conf.epp',  
      {  
        'group_name'=>$group_name,  
        'commands'=>$commands  
      }  
    ),  
    validate_cmd=>'/usr/sbin/visudo -cf %',  

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
file{"/etc/sudoers.d/${group_name}":
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0440',
  content=>epp('masterpract/sudoers.conf.epp',
    {
      'group_name'=>$group_name,
      'commands'=>$commands
    }
  ),
  validate_cmd=>'/usr/sbin/visudo -cf %',
  require=>Group[$group_name]
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/manifests$ sudo cat install.pp
class masterpract::install{

  package{'nginx':
    ensure=>installed
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/manifests$
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract$ cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/templates$ ls
index.html.epp  nginx.conf.epp  sudoers.conf.epp
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/templates$ sudo cat index.html.epp
<!DOCTYPE html>
<h1>Hi from Vishnu's webapp served from <%= $document_root %>
<h2>IP: <%= $facts['networking']['ip'] %> </h2>
<h2>OS: <%= $facts['os']['name'] %> </h2>
<h2>Hostname: <%= $facts['networking']['hostname'] %> </h2>

ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/masterpract/templates$ sudo cat nginx.conf.epp
server {

  listen <%= $port %>;
  server_name _;
  root <%= $document_root %>;
  index index.html;
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
  
access_log <%= $log_dir %>/access.log;  
error_log <%= $log_dir %>/error.log;  
  
location / {  
    try_files $uri $uri/ =404;  
}  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/templates$ sudo cat sudoers.conf.epp  
<%= $group_name %> ALL=(ALL) <%= $commands[0] %>  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/templates$ cd ..  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract$ cd functions  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/functions$ ls  
normalize_name.pp  user_home.pp  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/functions$ sudo cat normalize_name.pp  
function masterpract::normalize_name(String $name) >> String {  
    $name.downcase()  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
}  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/templates$ sudo cat sudoers.conf.epp  
<%= $group_name %> ALL=(ALL) <%= $commands[0] %>  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/templates$ cd ..  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract$ cd functions  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/functions$ ls  
normalize_name.pp  user_home.pp  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/functions$ sudo cat normalize_name.pp  
function masterpract::normalize_name(String $name) >> String {  
    $name.downcase()  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/functions$ sudo cat user_home.pp  
function masterpract::user_home(String $name) >> String {  
    "/home/${masterpract::normalize_name($name)}"  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/master  
pract/functions$
```

Agent:



Hi from Vishnu's webapp served from /var/www/webapp

IP: 172.31.9.17

OS: Ubuntu

Hostname: puppet-agent01

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshing CA certificate  
Info: CA certificate is unmodified, using existing CA certificate  
Info: Refreshing CRL  
Info: CRL is unmodified, using existing CRL  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Info: Loading facts  
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1788254443'  
Notice: /Stage[main]/Masterpract::Users/File[/home/hari/.ssh/authorized_keys]/content:  
--- /home/hari/.ssh/authorized_keys      2026-09-01 09:14:44.752196661 +0000  
+++ /tmp/puppet-file20260901-2136-ojwljq  2026-09-01 09:20:45.003995635 +0000
```

```

ubuntu@puppet-agent01: ~
Notice: /Stage[main]/Masterpract::Users/File[/home/hari/.ssh/authorized_keys]/content: content changed '{sha256}87a5ff69b252f78b2b1a7720bb1dc462ccb661ca9a92ef1359e9acdee3e6de54' to '{sha256}adc49fde5b1a16a651f9cd8cefdd5d21b5045711cdc59b6fef86f881a33d6f47'
Notice: /Stage[main]/Masterpract::Users/User[vishh]/ensure: created
Notice: /Stage[main]/Masterpract::Users/File[/home/vishh/.ssh]/ensure: created
Notice: /Stage[main]/Masterpract::Users/File[/home/vishh/.ssh/authorized_keys]/ensure: defined content as '{sha256}87a5ff69b252f78b2b1a7720bb1dc462ccb661ca9a92ef1359e9acdee3e6de54'
Notice: Applied catalog in 0.67 seconds
ubuntu@puppet-agent01:~$ curl http://localhost
<!DOCTYPE html>
<h1>Hi from Vishnu's webapp served from /var/www/webapp
<h2>IP: 172.31.9.17 </h2>
<h2>OS: Ubuntu </h2>
<h2>Hostname: puppet-agent01 </h2>

ubuntu@puppet-agent01:~$ getent group webapp
webapp:x:997:
ubuntu@puppet-agent01:~$ id vishh
uid=1003(vishh) gid=997(webapp) groups=997(webapp)
ubuntu@puppet-agent01:~$ id hari
uid=1002(hari) gid=997(webapp) groups=997(webapp)
ubuntu@puppet-agent01:~$ ls -ld /home/vishh /home/hari
drwxr-x--- 3 hari webapp 4096 Aug 20 08:32 /home/hari
drwxr-x--- 3 vishh webapp 4096 Sep  1 09:20 /home/vishh

```

```

ubuntu@puppet-agent01: ~
ubuntu@puppet-agent01:~$ ls -ld /home/vishh /home/hari
drwxr-x--- 3 hari webapp 4096 Aug 20 08:32 /home/hari
drwxr-x--- 3 vishh webapp 4096 Sep  1 09:20 /home/vishh
ubuntu@puppet-agent01:~$ ls -l /home/vishh/.ssh/authorized_keys
ls: cannot access '/home/vishh/.ssh/authorized_keys': Permission denied
ubuntu@puppet-agent01:~$ sudo ls -l /home/vishh/.ssh/authorized_keys
-rw----- 1 vishh webapp 22 Sep  1 09:20 /home/vishh/.ssh/authorized_keys
ubuntu@puppet-agent01:~$ sudo ls -l /home/hari/.ssh/authorized_keys
-rw----- 1 hari webapp 24 Sep  1 09:20 /home/hari/.ssh/authorized_keys
ubuntu@puppet-agent01:~$ sudo ls -l /etc/sudoers.d/webapp
-r--r----- 1 root root 50 Sep  1 09:18 /etc/sudoers.d/webapp
ubuntu@puppet-agent01:~$ sudo crontab -u www-data -l
# HEADER: This file was autogenerated at 2026-09-01 09:14:44 +0000 by puppet.
# HEADER: While it can still be managed manually, it is definitely not recommended.
# HEADER: Note particularly that the comments starting with 'Puppet Name' should not be deleted, as doing so could cause duplicate cron jobs.
# Puppet Name: log-cleanup
0 2 * * * /usr/bin/find /var/log/webapp -type f -mtime +7 -delete
ubuntu@puppet-agent01:~$ ls -l /etc/nginx/sites-enabled/
total 0
lrwxrwxrwx 1 root root 37 Sep  1 09:14 vishsite -> /etc/nginx/sites-available/nginx.conf
lrwxrwxrwx 1 root root 38 Aug 24 04:32 webapp.conf -> /etc/nginx/sites-available/webapp.conf
ubuntu@puppet-agent01:~$ |

```

```
ubuntu@puppet-agent01: ~
Info: Loading facts
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent01
Info: Applying configuration version '1788263609'
Notice: Applied catalog in 0.54 seconds
ubuntu@puppet-agent01:~$ sudo curl http://localhost
<!DOCTYPE html>
<h1>Hi from Vishnu's webapp served from /var/www/webapp
<h2>IP: 172.31.9.17 </h2>
<h2>OS: Ubuntu </h2>
<h2>Hostname: puppet-agent01 </h2>

ubuntu@puppet-agent01:~$ sudo ls -ld /var/log/webapp
drwxr-x--- 2 www-data www-data 4096 Sep  1 09:18 /var/log/webapp
ubuntu@puppet-agent01:~$ sudo ls -l /var/log/webapp/access.log
-rw-r--r-- 1 root root 0 Sep  1 09:18 /var/log/webapp/access.log
ubuntu@puppet-agent01:~$ sudo ls -l /var/log/webapp/error.log
-rw-r--r-- 1 root root 0 Sep  1 09:18 /var/log/webapp/error.log
ubuntu@puppet-agent01:~$ sudo cat /home/vishh/.ssh/authorized_keys | grep vishh
ssh-rsa AAAA...vishh-key
ubuntu@puppet-agent01:~$ sudo cat /home/hari/.ssh/authorized_keys | grep hari
ssh-rsa AAAA...hari-key
ubuntu@puppet-agent01:~$ |
```